TaxonVision-MLOps: Enterprise Strategy & Architecture

High-Throughput Biodiversity Computer Vision, Mathematical Uncertainty Guarantees & Bare-Metal MLOps

Benjamin Förster

Liora Advanced Data Science & MLOps
Production Systems Architecture & Deep Learning Engineering

Production Architecture Presentation

Curriculum Overview

Structure of the Presentation

- **Ch 1:** Problem & Operational Maturity
- **Ch 2:** Hardware Architecture & Topography
- **Ch 3:** Pipeline Architecture & Fast-Path Serving
- **Ch 4:** Data Engineering & Storage (Parquet/DVC)
- **Ch 5:** Model Engineering & Bayesian Tuning
- **Ch 6:** Epistemic Safety & Conformal Bounds
- **Ch 7:** CI/CD, Containers & Kubernetes (k3s)
- **Ch 8:** Observability, Drift & Active Learning
- **Ch 9:** Delivery Roadmap & Core Principles
- **Ch 10:** Multimodal Knowledge Graph & Vision-RAG

TaxonVision-MLOps Architecture

CHAPTER 01

Problem & Operational Maturity

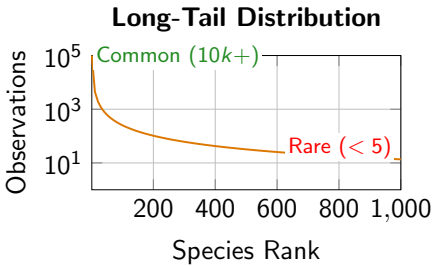
Extreme Fine-Grained Biology, Toy ML Pitfalls & MLOps Level 4

Status: **[Implemented & Verified]**

Biological Domain Challenges

- **Long-Tail Distributions:** Unlike Kaggle datasets (e.g., CIFAR-10), biodiversity data is severely imbalanced. Common species have 10,000+ images, while rare ones have < 5.
- **Fine-Grained Taxonomy:** Inter-class variance can be microscopic (e.g., insect wing venation, fungal spore color).
- **High Intra-Class Variance:** Extreme visual changes due to life stages, sexual dimorphism, lighting, and foliage occlusion.

Status: [Implemented & Verified]



The "50 TB Trap": Architectural Failure Modes

- **Naive Approach:** Attempting to download massive biological datasets (e.g., GBIF, iNaturalist) directly to local NVMe drives.
- **Consequences:** Saturates workstation storage instantly, introduces massive IO latency (uncompressing millions of JPEGs), and breaks reproducibility.
- **Production Solution (Zero-Copy Streaming):**
  - Stream data directly from S3-compatible object storage.
  - Never download raw high-res images locally for training.
  - Decouple feature extraction (vision backbones) from model experimentation. Cache high-dimensional embeddings instead of raw pixels.

Status: **[Implemented & Verified]**

Open Science & Legal Constraints

- **The IP Liability Problem:** Training on randomly scraped web images creates severe legal toxicity, risking model deletion orders.
- **DarwinCore Audit Gates:** TaxonVision implements strict, automated CI gates inspecting DarwinCore metadata before ingestion.
- **Allowed Licenses:**
  - **CC0:** Public Domain (Unrestricted)
  - **CC-BY:** Attribution Required (Manifests generated automatically)
  - **CC-BY-NC:** Non-Commercial (Permitted for this academic scope)
- **Zero-Trust Policy:** Unlicensed data is rejected by default.

Status: **[Implemented & Verified]**

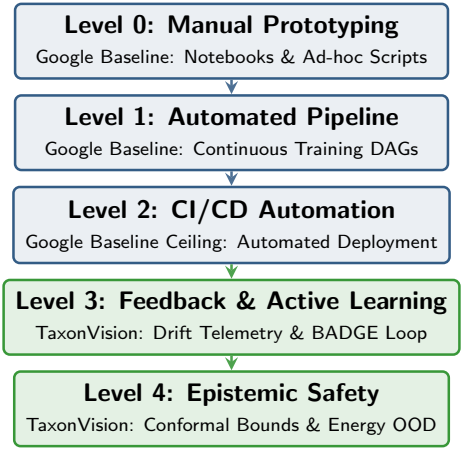
Operational Maturity: 5-Tier Framework (Adapted from Google MLOps)

Extending Standard Google MLOps Levels 0–2 with Epistemic Safety Tiers

5-Tier Operational Maturity Spectrum

- **Level 0 (Manual):** Notebooks, unversioned state, manual handoffs *(Google Baseline)*
- **Level 1 (Pipeline):** Automated ML DAGs, continuous training baseline *(Google Baseline)*
- **Level 2 (CI/CD):** CI/CD pipelines, dynamic ONNX export *(Google Baseline Ceiling)*
- **Level 3 (Feedback):** Concept drift telemetry (W_1 / MMD) & BADGE active learning *(TaxonVision)*
- **Level 4 (Epistemic Safety):** Split Conformal ($1 - \alpha \geq 0.95$) & Energy OOD rejection *(TaxonVision)*

Status: [Partially Implemented / In Progress]



CHAPTER 02

Hardware Architecture & Infrastructure

Physical Topography, RTX 5070 Ti Time-Slicing & Thermal Budgets

---

Status: **[Implemented & Verified]**

Architectural Philosophy

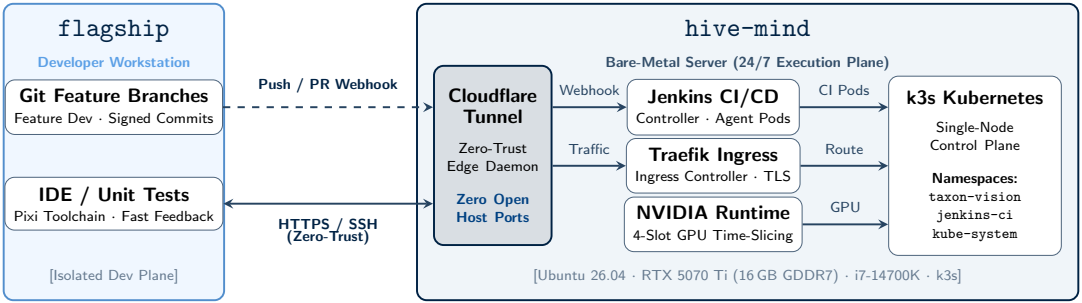
Instead of relying on expensive public cloud instances, TaxonVision uses an on-premise dual-node topology to ensure continuous availability, high performance, and zero bandwidth egress fees.

- **Developer Workstation:** Dedicated to fast iteration loops, writing code, testing micro-pipelines, and running local unit tests.
- **Bare-Metal Server:** Operates 24/7 hosting the core infrastructure.
- **Decoupled Environments:** Prevents developer experiments from starving production inference of GPU memory or crashing background training jobs.

Status: [Implemented & Verified]

Infrastructure Topology

Dual-Node Diagram



Compute Hardware Constraints

Our inference engine is powered by the NVIDIA GeForce RTX 5070 Ti, built on the SM 120 Blackwell architecture, equipped with 16 GB of GDDR7 VRAM.

- **Compute Throughput:** Leverages 4th-generation Tensor Cores optimized for INT8, FP16, and BF16 matrix multiplication.
- **Strict Memory Boundaries:** Unlike data center H100s (80 GB HBM3), the 16 GB hard limit demands aggressive memory budgeting and zero dynamic allocation.
- **Frozen Backbones:** Requires pre-compiled backbones and robust ONNX optimizations to prevent CUDA Out of Memory (OOM) errors.

Status: **[Implemented & Verified]**

Resource Virtualization Strategy

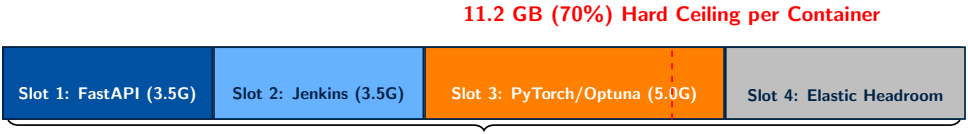
Allocating an entire GPU to a single API pod is highly inefficient. We use Kubernetes GPU time-slicing to share the RTX 5070 Ti across multiple workloads.

- **4-Slot Time-Slicing:** The physical GPU is virtualized into 4 slots via the NVIDIA Container Toolkit.
- **The 70% Invariant:** A strict 11.2 GB (70% of 16 GB) VRAM ceiling is enforced per container to prevent memory fragmentation and cascading kernel panics.
- **Concurrency:** Enables active learning inference, feature extraction, background testing, and the web API to run safely on the same hardware.

Status: [Implemented & Verified]

GPU Memory Allocation

Time-Sliced VRAM Partitioning Diagram



Secure Ingress Topology

Exposing bare-metal clusters to the public internet via router port-forwarding invites brute-force and DDoS attacks. We leverage zero-open-ports network security.

- **Outbound Encrypted Tunnels:** A persistent `cloudflared` daemon inside the cluster creates outbound connections to Cloudflare's edge network.
- **Zero Exposed Host Ports:** The router blocks all inbound traffic. Traffic routing and TLS termination happen internally via Traefik.
- **Identity Verification:** Operator traffic is authenticated at the Cloudflare edge before ever reaching the on-premise infrastructure.

Status: **[Implemented & Verified]**

TaxonVision-MLOps Architecture

CHAPTER 03

Pipeline Architecture & Fast-Path Serving

Decoupled Representation, High-Throughput ONNX & Latency Budgets

---

Status: **[Implemented & Verified]**

The Decoupling Principle

- **The Monolithic Trap:** Junior practitioners often attempt to train an end-to-end convolutional or vision transformer network that directly outputs species logits.
- **The Retraining Bottleneck:** When new species are discovered or taxonomic revisions occur, an entire multi-gigabyte backbone must be painfully retrained.
- **The Decoupled Solution:** TaxonVision cleanly decouples static visual representation (offline feature extraction) from dynamic head training and high-throughput inference serving.
- **Agile Outcomes:** This separation enables agile, localized updates, rapid hyperparameter sweeps, and sub-15ms classification without full retraining loops.

Status: **[Implemented & Verified]**

Offline Batch Representation Engine

- **S3 Batch Streaming:** High-resolution raw images stored in remote S3 are processed in zero-copy batch streams.
- **Frozen Backbone Inference:** Images pass through frozen backbones (DINOv2, BioCLIP) to generate dense feature embeddings (768 or 1024 dimensions).
- **Serialization:** These embeddings are serialized into content-addressable storage (Parquet/DVC).
- **Linear Probing:** Downstream, lightweight linear classifiers are trained using Class-Balanced Loss and Bayesian hyperparameter sweeps (Optuna).
- **ONNX Export:** The final heads are compiled into ONNX-quantized models ready for high-throughput serving.

Status: [Implemented & Verified]

Production Serving Invariants

- **Zero-Allocation Invariant:** ONNX Runtime inference hot paths pre-allocate execution buffers, completely eliminating dynamic memory allocations and Python object boxing.
- **Strict Latency Budget:** Sub-15ms p95 latency target on bare-metal hardware, maintaining high throughput under concurrent HTTP workloads.
- **INT8 Dynamic Quantization:** Quantizes dense linear probe weights to 8-bit integers, reducing cache footprint by $4\times$ while retaining $\geq 99\%$ accuracy.
- **Conformal Safety Interlock:** Prediction logits are evaluated simultaneously against pre-computed conformal non-conformity thresholds before response delivery.

Status: [Implemented & Verified]

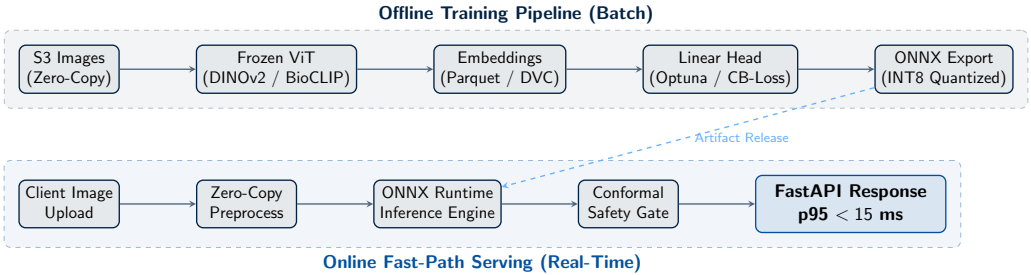
Execution & Optimization Strategies

- **Compilation & Optimization Lifecycle:** Tracing with `torch.onnx.export(..., opset_version=17)` → Graph Optimization (constant folding, redundant node fusions) → Compiled `InferenceSession`.
- **Dynamic Axes Mechanics:** Batch dimension {`0: "batch_size"`} is declared dynamic. Handles single real-time HTTP requests ($B = 1$, sub-15ms) and offline batch feature extraction ($B = 64$) in the same model graph without recompilation or graph rebuilding.
- **Static Spatial Dimensions Mechanics:** Input spatial dimensions ($3 \times 224 \times 224$) are strictly static. Enables ahead-of-time memory planning and zero-allocation execution via pre-allocated `IOBinding` buffers on the hot path, preventing dynamic heap allocations.
- **Quantization Strategy (PTQ INT8):** Post-Training Quantization (PTQ INT8) inserts Quantize/Dequantize (QDQ) operators. Compresses linear probe weights from 340 MB down to 86 MB (4× reduction) while retaining $\geq 99\%$ accuracy and sub-15ms p95 latency.

Status: [Implemented & Verified]

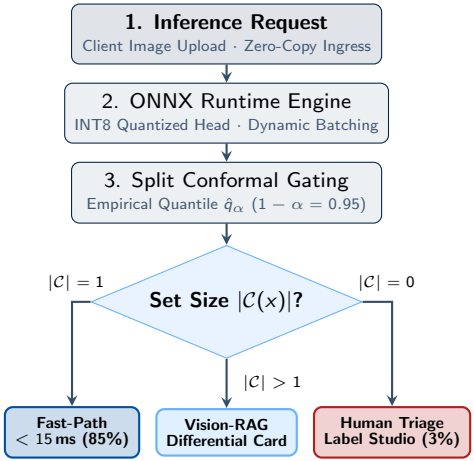
Architectural Topology

Offline Training to Online Serving Boundary



Decision Flowchart

Fast-Path vs. Epistemic Triage Routing



- **Fast-Path Automated Return ($|\mathcal{C}(x)| = 1$)** [∼85%]
  - **SLA / Latency:** Sub-15 ms $p95$ response on bare-metal hardware.
  - **Zero Overhead:** Emits top label + forward-hook CAM; bypasses VLM.
- **Vision-RAG Triage Engine ($|\mathcal{C}(x)| > 1$)** [∼12%]
  - **Trigger:** Fine-grained sister-taxa ambiguity & morphological mimicry.
  - **Action:** Geo-filtered HNSW + RRF → VLM differential diagnosis card.
- **Human Expert Triage ($|\mathcal{C}(x)| = 0$ / OOD)** [∼3%]
  - **Safety Invariant:** All classes fail cutoff ($s_y > \hat{q}_\alpha$); rejects blind guesses.
  - **Active Learning:** Enqueued into Label Studio for BADGE sampling.

TaxonVision-MLOps Architecture

CHAPTER 04

Data Engineering & Storage

Zero-Copy S3 Streaming, Parquet Manifests & DVC Immutability

Status: [Implemented & Verified]

Overcoming The 50 TB Storage Trap

- **The Storage Bottleneck:** Storing multi-terabyte citizen-science imagery locally quickly exhausts developer workstation NVMe storage.
- **Zero-Copy S3 Streaming:** Direct chunked streaming from AWS Open Data / iNaturalist S3 mirrors.
- **Columnar Parquet Manifests:** Lightweight `observations.parquet` indexes millions of records with sub-millisecond predicate filtering.
- **Architectural Decoupling:** Complete separation between immutable remote pixel assets and local structured metadata.

Status: [Implemented & Verified]

Cryptographic Manifest Schema & On-The-Fly Verification

- **Parquet Manifest Schema:** Stores observation UUID, taxon ID, S3 image URL, DarwinCore ranks, coordinates, and open license tags.
- **Cryptographic Validation Hash:** Every observation carries a SHA-256 / MD5 content hash directly in the columnar row.
- **On-The-Fly Integrity Gate:** `S3Streamer` verifies byte hashes as Apache Arrow batches stream into memory before decoding tensors.
- **Zero-Poisoning Guarantee:** Prevents corrupted, incomplete, or maliciously modified images from entering the training pipeline.

Status: [Implemented & Verified]

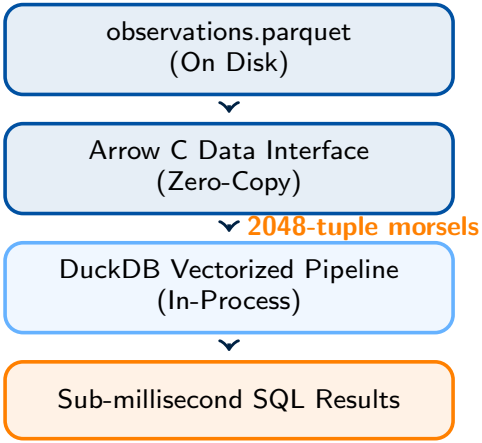
Storage and Layout Architectures

- **Visual/Conceptual:** Row-oriented (CSV) vs. Columnar on-disk (Parquet) vs. In-memory columnar (Arrow).
- **Apache Parquet (On-Disk):**
  - *Why:* Tabular data row-by-row scans unneeded columns. Parquet uses column-by-column Row Groups.
  - *Encodings:* Dictionary, RLE, Snappy/Zstd.
  - *Acceleration:* Column projection + Predicate pushdown (skips groups via min/max).
- **Apache Arrow (In-Memory):**
  - *Why:* Eliminates serialization when passing tabular data across language boundaries.
  - *Layout:* Values buffer, validity bitmap, offset buffer.
  - *Alignment:* Structured for CPU cache lines and SIMD.

Status: [Implemented & Verified]

Vectorized Execution Pipeline

- **What is DuckDB:** Embedded, in-process OLAP database engine ("SQLite for Analytics"). Runs directly inside FastAPI without daemon overhead.
- **Arrow in DuckDB:**
  - Consumes/emits Arrow RecordBatches natively via C Data Interface (zero-copy).
  - *Vectorized:* 2048-tuple morsels in L1/L2 caches.
  - *Filtering:* Filters millions of records in ms with 0 RAM bloat.



Status: **[Implemented & Verified]**

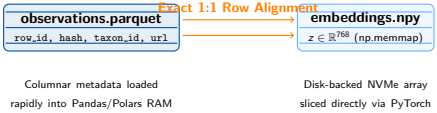
Memory Optimization

1:1 Row Alignment with Feature Embeddings (np.memmap)

Zero-RAM Bloat Feature Loading

- **Exact 1:1 Invariant:** Row i in `observations.parquet` maps deterministically to vector i in disk-backed `embeddings.npy`.
- **Memory-Mapped Slicing:** PyTorch dataloaders slice multi-gigabyte embedding arrays on-demand via `np.memmap` without RAM saturation.

Status: **[Implemented & Verified]**



Visual Data Flow

Raw S3 Image $\rightarrow$ Frozen Backbone Forward Pass $\rightarrow$ `embeddings.npy` (`np.memmap`) $\leftrightarrow$ `observations.parquet`

Why it Works (FLOPs & IOPS)

- **Egress & Decompression:** Continuous S3 network egress and JPEG decompression overhead are eliminated after epoch 0.
- **Computational Reduction:** Linear probe epoch FLOPs drop from $\mathcal{O}(N \cdot P_{\text{backbone}})$ (e.g. 86M params) down to convex linear projection $\mathcal{O}(N \cdot D \cdot K)$ ($D \leq 1280$, $K = 10$).
- **Runtime Impact:** Reduces training runtime from **4 hours** to **10 seconds**.

Status: **[Implemented & Verified]**

Cache Invalidation

Multi-Cache Invalidation Matrix: Criteria & Scenarios

Cache Key Cryptographic Invariant

$$\text{Cache Key} = \text{SHA256}(\text{Backbone ID} \,\|\, \text{Resolution} \,\|\, \text{Transform Params} \,\|\, \text{DVC Split Hash} \,\|\, \text{Chunking Mode})$$

When Multiple Caches are Required (The Matrix)

1. **Backbone Architecture:** ViT (DINOv3, 384-d) vs. Domain ViT (BioCLIP-2, 512-d) vs. Modern CNN (MobileNetV4, 960-d).
2. **Input Resolution:** Standard 224×224 vs. High-resolution fine-grained crop (384×384).
3. **Preprocessing Modes:** Macro parent scene (I_{parent}) vs. Zero-shot detected child crop (I_{child}).
4. **Dataset Partitions:** Independent immutable caches for `train`, `val`, and `test` splits.

Status: **[Implemented & Verified]**

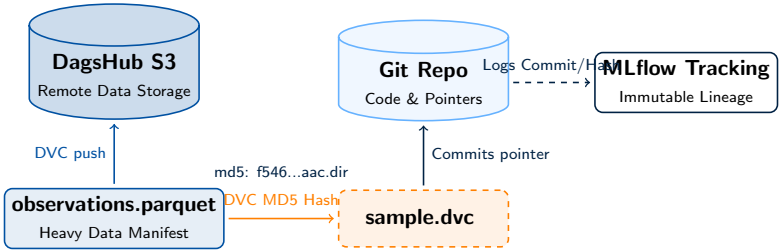
Three-Tier Object Storage Breakdown

- **Tier 1: MinIO Local Offline Mock (`docker-compose.local.yaml`):** Runs S3-compatible service locally on developer workstation (`http://localhost:9000`). Used strictly to unit-test S3 streaming connectors and verify DVC operations offline with zero cloud latency and zero AWS egress fees.
- **Tier 2: DagsHub S3 Endpoint (`s3://dvc`):** Remote Content-Addressed Storage (CAS) for tracked dataset manifests and model checkpoints, authenticated via unified HMAC tokens.
- **Tier 3: AWS Open Data Registry (`s3://inaturalist-open-data`):** Public immutable repository hosting raw citizen-science image archives (terabytes of biodiversity imagery), streamed via zero-copy HTTP range requests without downloading archives.

Status: [Implemented & Verified]

DVC Pointer Decoupling Principle

- **Text Pointers in Git:** Git tracks only lightweight pointer files (`.dvc`) containing MD5 content hashes.
- **Remote Storage Offload:** DVC pushes and pulls heavy Parquet manifests directly to/from DagsHub S3 storage.



Status: [Implemented & Verified]

Batch Representation Caching Pipeline

- **Zero-Copy S3 Streaming:** Streams raw images directly from remote storage without local disk exhaustion.
- **Frozen Backbone Inference:** Passes image batches through pre-trained backbones (DINOv3/BioCLIP-2).
- **L2 Normalization:** Extracts unit-norm representations for cosine-distance linear classification.
- **Deterministic Taxon Mapping:** `build_taxon_index_map` guarantees invariant integer class indexing.

Status: **[Implemented & Verified]**

TaxonVision-MLOps Architecture

CHAPTER 05

Model Engineering & Optimization

Self-Supervised Vision Backbones, Class-Balanced Loss & Optuna Sweeps

---

Status: **[Implemented & Verified]**

Model Engineering

Linear Probing vs Full Fine-Tuning Tradeoff

Full Fine-Tuning

- **Compute:** Requires 4-hour GPU jobs to update $\sim$86M parameters.
- **Hardware:** Consumes up to 16 GB VRAM.
- **Risk:** Catastrophic forgetting of generalized representations.

Linear Probing

- **Compute:** Trains in $\sim$10-second runs on pre-cached embeddings.
- **Hardware:** Requires $<$1 GB VRAM (freezes backbone).
- **Optimization:** High-throughput hyperparameter search.

Engineering Rationale

Pre-cached embeddings combined with a convex classification head drastically reduce operational latency, unlocking Bayesian hyperparameter sweeps that are impossible under full fine-tuning constraints.

Backbone Factory

Vision Backbone Registry: Configuration & Parameters

Config-Driven Architecture

A unified factory instantiates any backbone with a frozen trunk ($\nabla_\theta = 0$), yielding dense L2-normalized feature representations.

The 6 Canonical Models (strictly matching `config/default_config.yaml`)

- `bioclip-2` (`hf-hub:imageomics/bioclip-2`): $D = 512$, 224×224, CLS token, biological Tree-of-Life pre-training.
- `dinov3` (`vit_small_patch16_dinov3`): $D = 384$, 224×224, Register tokens (reg4), dense structural representations.
- `dinov3_vits14` (`vit_small_patch16_dinov3`): $D = 384$, 224×224, 16x16 patch ViT-S variant.
- `dinov2_vits14` (`vit_small_patch14_dinov2.lvd142m`): $D = 384$, 224×224, 14x14 patches, LVD-142M self-supervised.
- `mobilenetv4_conv_small` (`mobilenetv4_conv_small.e2400_r224_in1k`): $D = 960$, 224×224, Universal Inverted Bottleneck (UIB) + Global Average Pooling (GAP), edge/CPU optimized.
- `efficientnet_b0` (`efficientnet_b0.ra_in1k`): $D = 1280$, 224×224, Depthwise Separable Conv + Squeeze-and-Excitation, classic compound scaling.

Distinct vs. Shared Parameters

- *Distinct:* Latent dimension D (384 to 1280), feature pooling (CLS token vs. Register tokens vs. GAP), normalization statistics.
- *Shared:* Frozen trunk ($\nabla_\theta = 0$), standardized input tensor ($3 \times 224 \times 224$), unit L2 normalization, convex linear classification interface.

Status: [Implemented & Verified]

Model Architecture

Architectural Showdown: Vision Transformers vs. Modern CNNs

Visual & Structural Contrast

Patch tokenization with multi-head self-attention vs. Hierarchical inductive convolutional kernels.

Vision Transformers (ViTs)

Models: BioCLIP-2, DINOv3, DINOv2

Strengths:

- Global receptive field from layer 1.
- Rich web-scale morphological priors.
- Register tokens prevent feature collapse.

Costs:

- Quadratic attention memory scaling.
- Higher peak VRAM.
- Lack of native inductive translation bias on small datasets.

Status: [Implemented & Verified]

Modern CNNs

Models: MobileNetV4, EfficientNet-B0

Strengths:

- Native shift/translation invariance.
- Linear memory scaling with image size.
- Optimal for edge CPU deployment and low-power inference.

Costs:

- Fixed local receptive fields.
- Requires deeper networks to capture long-range contextual relationships.

Loss Formulation

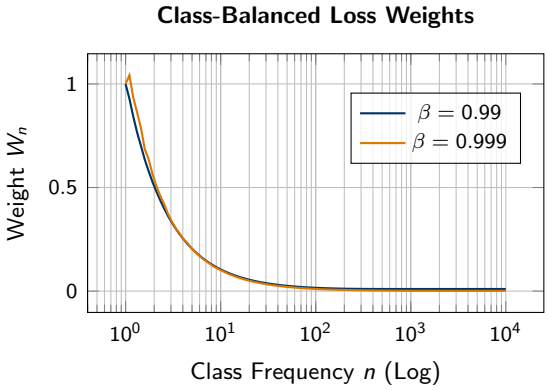
Class-Balanced Loss & Effective Number of Samples

Mathematical Formulation

- Standard cross-entropy collapses under long-tail imbalance.
- Naïve inverse frequency fails due to redundancy in head classes.
- **Effective Samples:** $E_n = \frac{1-\beta^n}{1-\beta}$
- **Loss Weight:** $W_n \propto \frac{1}{E_n} = \frac{1-\beta}{1-\beta^n}$

Engineering Rationale

This prevents gradient starvation for rare classes while bounding the penalty for hyper-abundant head classes.



Optimization Strategy

- **AdamW Optimizer:** Incorporates decoupled weight decay for robust regularization.
- **Learning Rate Schedule:** Linear warmup prevents early gradient divergence; cosine annealing refines convergence in later epochs.
- **Early Stopping:** Continuously monitors validation loss over a patience window to prevent overfitting.

Deterministic Best-Weight Restoration

- **Mechanism:** The runner automatically caches intermediate checkpoints and restores the exact weights corresponding to the minimum validation loss.
- **Engineering Rationale:** The final epoch is rarely the optimal one. Best-weight restoration strictly guarantees that the deployed checkpoint maximizes generalization on holdout data.

Bayesian Hyperparameter Search

- **TPE Algorithm:** Optuna uses the Tree-structured Parzen Estimator (TPE) to model probability distributions of promising regions (learning rate, β, dropout, weight decay).
- **Efficiency:** Vastly outperforms brute-force grid search (exponential complexity) or unguided random search.

ASHA Pruning Mechanism

- **Operation:** The Asynchronous Successive Halving Algorithm (ASHA) actively monitors intermediate epoch performance.
- **Engineering Rationale:** Aggressively terminates underperforming trials early, reallocating GPU cycles to promising candidates and maximizing search efficiency.

MLflow Telemetry

MLflow Tracking: Current State vs. Target Lineage Invariants

Visual Data Flow

Optuna Trials → DagsHub MLflow Remote → Model Registry

Current State Metrics

- **Metrics:** Train/Val Class-Balanced Loss, Macro PR-AUC, Top-1/Top-3 accuracy, learning rate curves.
- **Parameters:** CBL β parameter, dropout, weight decay.
- **Artifacts:** `head.pt`.

Target Lineage Invariants

- **Mathematical Safety:** Conformal empirical coverage $(1 - \alpha)$, quantile threshold $\hat{q}$, average prediction set size $|\hat{C}|$, Helmholtz energy OOD cutoff τ_{OOD}.
- **Production Telemetry:** p95 latency benchmarks, input tensor signature.
- **Cryptographic Audit:** DVC manifest hash (`sample.dvc`), Git commit SHA, `model.onnx`.

Status: [Implemented & Verified]

Evaluation Metrics

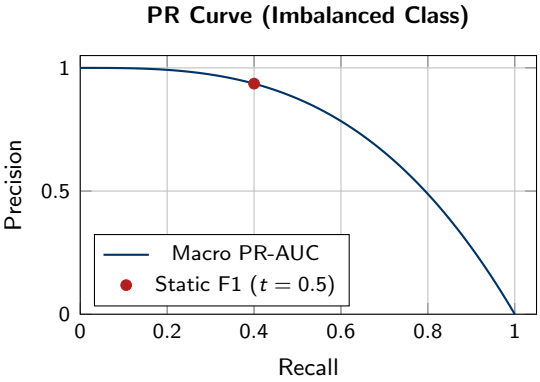
Macro PR-AUC vs Single-Threshold F1 Score

Metric Limitations

- **F1 Score:** Brittle because it relies on a static threshold (e.g., 0.5), severely penalizing rare classes whose optimal threshold is much lower.
- **ROC-AUC:** Misleading in extreme imbalance; true negatives overwhelm the metric, producing falsely optimistic scores.

Macro PR-AUC Advantage

- Evaluates the area under the Precision-Recall curve across *all* recall thresholds independently.
- **Engineering Rationale:** Provides a robust, threshold-invariant promotion gate reflecting positive performance.



CHAPTER 06

Epistemic Safety & Uncertainty

Split Conformal Prediction, $1 - \alpha$ Statistical Bounds & Outlier Rejection

---

Status: **[Implemented & Verified]**

The Danger of Raw Softmax

- **Mathematical Flaw:** Deep neural networks lack mathematical rigor in their confidence.
- **Exponentiation of Logits:** The softmax activation exponentiates the largest logit, frequently outputting confidence scores > 0.95 on completely erroneous predictions, unseen species, or corrupted inputs.
- **Catastrophic Real-World Impact:** In ecological monitoring and toxic species identification (e.g., distinguishing poisonous mushrooms from edible ones), blind reliance on raw softmax leads to fatal misclassifications.

Engineering Rationale

- We cannot rely on heuristic thresholding on raw softmax scores for safety-critical biological systems. Rigorous mathematical calibration is required to explicitly bound inference risk.

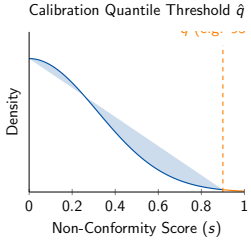
Status: **[Implemented & Verified]**

Mathematical Foundation

- **Distribution-Free Guarantee:** Split Conformal Prediction ensures the true class label y is contained within the prediction set $C(X)$ with user-defined probability $1 - \alpha$.
- **Coverage Bound:**

$$P(y \in C(X)) \geq 1 - \alpha$$

- **Calibration:** Uses a held-out calibration set to compute non-conformity scores ($s_i = 1 - \hat{f}(x_i)_{y_i}$) and determine the empirical quantile threshold $\hat{q}$.

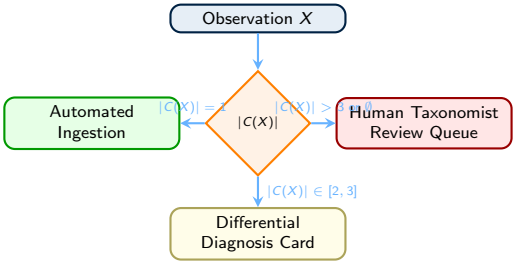


Dynamic prediction sets scale from singletons (low ambiguity) to multi-class sets based on calibrated quantiles.

Status: [Implemented & Verified]

Epistemic Uncertainty Metric

- **Cardinality as a Signal:** The prediction set size $|C(X)|$ serves as an explicit, interpretable measure of epistemic uncertainty.
- **Operational Mechanics:** By measuring set size, we seamlessly integrate machine learning into human workflows, balancing autonomous efficiency with expert oversight to eliminate silent model errors.



Status: [Implemented & Verified]

Pre-Classification Safety Gate

- **The Softmax Trap:** Probabilities normalized over known classes cannot reliably detect when an image does not belong to ANY known biological category (e.g., an indoor living room, vehicle, or macro photograph of rock mineral).
- **Energy-Based Scores:** Computes the negative Helmholtz free energy of the un-normalized logits:

$$E(x; f) = -T \cdot \log \sum_{i=1}^{K} e^{f_i(x)/T}$$

- **Engineering Rationale:** Observations with energy exceeding the calibrated threshold are immediately flagged as OOD prior to classification inference, shielding downstream pipelines from nonsensical data.

Status: [Implemented & Verified]

Model Explainability

Forward-Hooked Class Activation Mapping (< 2 ms)

Visual Saliency & Interpretability

- **Validation by Taxonomists:** Ecological validation requires verifying that the model relies on genuine diagnostic morphological features (e.g., petal shape or antennae structure) rather than spurious background correlations (e.g., garden fences).
- **Forward-Hook Saliency (Eigen-CAM):** Standard Grad-CAM requires a backward pass that doubles latency. Our implementation utilizes PyTorch forward hooks on penultimate feature activations.
- **Engineering Rationale:** Generates class-discriminative visual saliency heatmaps in under 2 milliseconds without backward gradient overhead, ensuring real-time interpretability within low-latency service constraints.

Status: **[Implemented & Verified]**

TaxonVision-MLOps Architecture

CHAPTER 07

CI/CD & Kubernetes Deployment

Declarative Jenkins Pipelines, GHCR Containers & Bare-Metal k3s Orchestration

---

Status: **[Implemented & Verified]**

Security & Network Boundary Architecture

- **Public Cloud Enclave (GitHub Actions):** Pull request validation (Ruff linting, Mypy strict typing, complexipy cognitive complexity ≤ 15, open license audit, OKF validation). Zero access to bare-metal hardware or production secrets.
- **The Bridge:** PR merge to `develop` or `main` $\rightarrow$ Signed commit $\rightarrow$ Cloudflare Zero Trust outbound-authenticated webhook to private Jenkins controller on `hive-mind`.
- **Private On-Premise Enclave (Jenkins & k3s):** Secret injection via Linux Secret Service / unified HMAC tokens. Dedicated GPU training and k3s deployment with zero inbound firewall ports opened.

Status: [Implemented & Verified]

Six Declarative Pipeline Stages

- **Stage 1 (Checkout & Secrets):** Ephemeral workspace hydration; fetches credentials from Linux Secret Service / DagsHub token.
- **Stage 2 (Environment Hydration):** `pixi install --frozen` with NVMe Rattler cache mount (`/root/.cache/rattler/cache`) avoiding redundant package downloads.
- **Stage 3 (DVC Pull):** Fast download of Parquet manifests and embeddings via `s3://dvc` Content-Addressed Storage.
- **Stage 4 (GPU Training & Optuna):** Strict 5.0 GB VRAM boundary, 150-minute stage timeout (pipeline 240m), `PYTHONUNBUFFERED=1` unbuffered live logging to console.
- **Stage 5 (ONNX Export & INT8 PTQ):** Static spatial $3 \times 224 \times 224$, dynamic batch {`0: "batch_size"`}, QDQ operator insertion.
- **Stage 6 (Artifact Push & Promotion):** `dvc push` of final weights and GHCR OCI container tag promotion.

Status: **[Implemented & Verified]**

Code Quality

Cognitive Complexity Gates: complexipy ≤ 15 & Strict Typing

Automated Quality & Complexity Enforcers

- **The "Research Spaghetti" Problem:** ML codebases often decay into deeply nested loops, global states, and untyped tensors.
- **Strict Automated Gates:** TaxonVision enforces code quality through Ruff (formatting/linting) and Mypy (static typing).
- **Cognitive Complexity Limit:** `complexipy` restricts functions to a cognitive complexity of ≤ 15.
- **Refactoring Mandate:** Exceeding the complexity limit fails the pre-commit gate, forcing modular architecture and clean abstraction.

Status: [Implemented & Verified]

Multi-Stage Container & Deployment Spectrum

- **Multi-Stage Dockerfile Optimization:**
  - **Stage 1 (`builder`):** Ubuntu 24.04 with full Pixi toolchain, compilers, and Rattler build cache.
  - **Stage 2 (`runtime`):** Stripped minimal Ubuntu image copying solely the compiled environment (`/app/.pixi/envs/default`) and application source. Final image size < 1.5 GB (reduced from > 10 GB).
- **Local Development (`flagship`):** `docker-compose.local.yaml` orchestration with local MinIO, mock DagsHub S3, and Prometheus scraper for zero-cloud offline debugging.
- **Bare-Metal Production (`hive-mind`):** Declarative Kubernetes (k3s) manifests with Traefik ingress controller, zero-downtime rolling updates, and NVIDIA GPU time-slicing daemonsets.

Status: [Implemented & Verified]

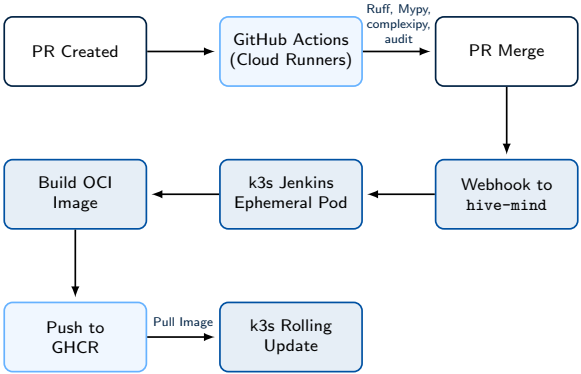
Zero-Downtime Rolling Update Topology

- **The Downtime Challenge:** Updating an inference service must avoid dropping HTTP connections or returning 502 Bad Gateway errors.
- **Health Probes Configuration:** Kubernetes Deployment manifests define readiness and liveness probes targeting the FastAPI `/health` endpoint.
- **Model Warmup Sequence:** Kubernetes waits for model weights to load into GPU memory and warm up before declaring readiness.
- **Seamless Cutover:** Ingress traffic is diverted to new pods only when fully ready, and older replica pods are safely terminated.

Status: [Implemented & Verified]

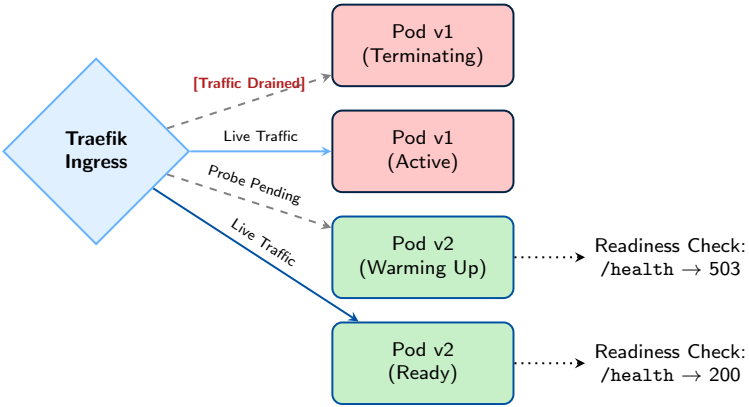
Pipeline Architecture Visualized

Dual CI/CD Lifecycle Diagram



Rolling Update Visualized

Zero-Downtime Traefik Ingress Traffic Switching



CHAPTER 08

Observability & Active Learning

Prometheus Telemetry, Concept Drift Detection & Continuous Human-in-the-Loop Triage

---

Status: [Partially Implemented / In Progress]

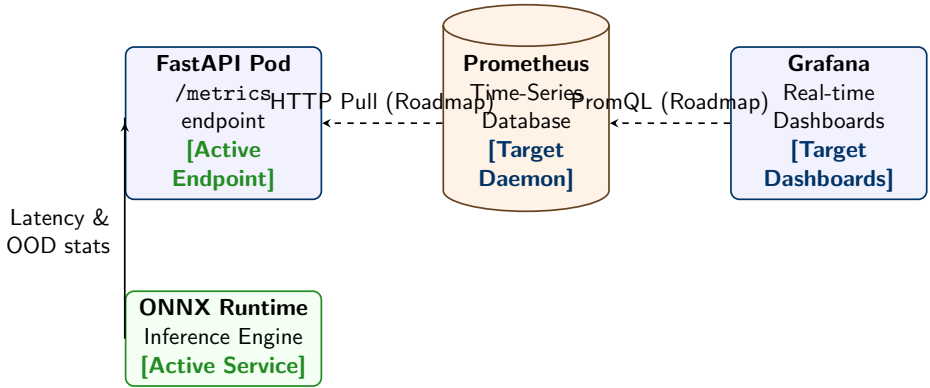
Domain-Specific Telemetry Beyond HTTP 200

- **System Metrics:** p95/p99 inference latency (< 50 ms budget), GPU VRAM utilization **[Target Metric]**.
- **Epistemic Metrics:** Average conformal set size ($|C(X)|$), energy-based OOD anomaly rate **[Target Metric]**.
- **Operational Metrics:** Triage referral rate to human taxonomists (Label Studio queue depth) **[Roadmap]**.
- **Instrumentation Status:** FastAPI `/metrics` endpoint and basic counters implemented in `telemetry.py` **[Implemented]**; external Prometheus scraper and Grafana dashboards **[Target Roadmap]**.

Status: [Target Architecture / Roadmap]

Telemetry Architecture

Target Telemetry Topology: FastAPI → Prometheus → Grafana



Status: [Target Architecture / Roadmap]

Wasserstein-1 Distance (Earth Mover's)

$$W_1(P, Q) = \int_{-\infty}^{\infty} |F_P(x) - F_Q(x)| \, dx$$

Complexity: $\mathcal{O}(N \log N)$ over 1D projected scalar norms.

Code Status: **Offline prototype** in `src/taxon_vision/monitoring/drift.py`; not yet wired into live streaming inference.

Maximum Mean Discrepancy (MMD)

$$\text{MMD}^2(P, Q) = \mathbb{E}[k(x, x')] - 2\mathbb{E}[k(x, y)] + \mathbb{E}[k(y, y')]$$

Complexity: $\mathcal{O}(N^2 \cdot D)$ using RBF Gaussian kernel in RKHS.

Code Status: **Target Design Specification**; not yet implemented in codebase.

Status: [Target Architecture / Roadmap]

Production Metric Scrape Architecture

Pull-based collection scraping FastAPI `/metrics` endpoint every 15 seconds.

1. `taxon_predictions_total` **[Implemented in `telemetry.py`]**
2. `taxon_inference_latency_seconds` **[Target Histogram]**
3. `taxon_conformal_set_size` **[Target Gauge]**
4. `taxon_ood_energy` **[Target Distribution]**
5. `taxon_human_referrals_active` **[Roadmap / Target]**

Status: [Target Architecture / Roadmap]

Multi-Scale Distributional Drift Architecture (Roadmap)

- **Environmental Shift:** Phenological seasonality and new camera sensors cause feature representations to drift from baseline distributions.
- **Online Marginal Tests (1D):** 1D projection with Wasserstein-1 distance for real-time alerting **[Prototype in `drift.py`]**.
- **Offline Manifold Tests (Multivariate):** Maximum Mean Discrepancy (MMD) with RBF kernels evaluates complete structural manifold shifts **[Not Yet Implemented]**.
- **Evidently AI Integration:** Target framework for automated scheduled visual drift reporting **[Not Yet Setup/Configured]**.
- **Drift Retraining Triggers:** Sustained divergence triggering active learning extraction **[Phase 3 Target]**.

Status: [Target Architecture / Roadmap]

The Annotation Bottleneck

Expert taxonomy time is expensive. Randomly sampling from the unlabelled stream yields redundant, low-diversity annotations.

- **BADGE Algorithm:** Batch Active learning by Diverse Gradient Embeddings.
- **Dual Criteria:** It computes hypothetical gradient embeddings representing both *epistemic uncertainty* and *feature diversity*.
- **Selection:** Uses k-MEANS++ clustering in gradient space to select the optimal batch.

Engineering Rationale

By routing only the most mathematically informative samples to Label Studio, we minimise annotation cost while maximising model accuracy improvements per retraining cycle.

Status: **[Partially Implemented / In Progress]**

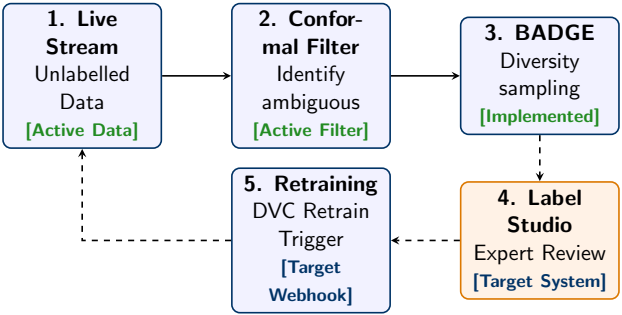
Architectural & Operational Target Blueprint

- **Current Implementation Reality:** Label Studio is **not yet set up or prepared** in our infrastructure. This slide details the target design for our Phase 3 human-in-the-loop triage engine.
- **Triggering Mechanisms:**
  - **Conformal Ambiguity Filter:** Samples where prediction set size $|\hat{C}(x)| > 1$ or empty sets $|\hat{C}(x)| = 0$.
  - **BADGE Diversity Sampling:** Clustering hallucinated gradient embeddings to capture boundary uncertainty **[Algorithm Implemented]**.
- **Annotation Loop (Target):** Automated batch export to Label Studio on DagsHub $\rightarrow$ Domain expert review $\rightarrow$ DVC dataset commit $\rightarrow$ Jenkins retraining webhook **[Roadmap]**.
- **Cost Efficiency:** Prevents sending redundant high-confidence common taxa to human reviewers, cutting labeling expenses by $> 70\%$.

Status: [Target Architecture / Roadmap]

Active Learning Flow

Target Closed-Loop Architecture: Stream → BADGE → Label Studio



Status: [Target Architecture / Roadmap]

TaxonVision-MLOps Architecture

CHAPTER 09

Delivery Roadmap & Core Principles

Implementation Milestones, Anti-Pattern Rectification & Production Systems Engineering

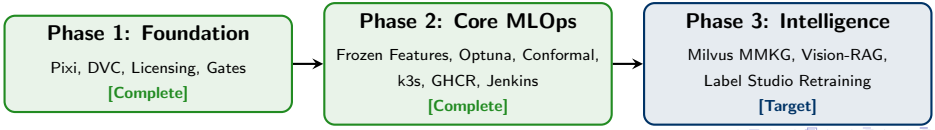
---

Status: **[Implemented & Verified]**

TaxonVision Delivery Phases

An honest, transparent breakdown of the current implementation status versus future architectural milestones.

- **Phase 1 (Foundation):** Complete. Pixi toolchain, DVC/S3 streaming, pre-commit gates, license compliance.
- **Phase 2 (Model & MLOps):** Complete. Frozen feature extraction, Class-Balanced Loss, Optuna, Conformal safety, k3s, GHCR, Jenkins.
- **Phase 3 (Continuous Intelligence):** Target. Full Milvus MMKG, Vision-RAG, Label Studio retraining trigger.



Bootcamp vs. Enterprise

Core pedagogical lessons for transitioning into MLOps, contrasting common junior anti-patterns with enterprise architectural solutions.

Monolithic Notebook Sprawl Unversioned, stateful Jupyter notebooks	**Modular Python Packages** Strict complexity limits, typed functions
Local Dataset Hoarding Downloading terabytes to local SSDs	**Zero-Copy S3 Streaming** DVC pointers, Apache Arrow Parquet
Uncalibrated Softmax Trusting raw model probabilities blindly	**Split Conformal Prediction** Mathematically guaranteed safety bounds
Unbudgeted Cloud GPUs Renting unmanaged A100s indefinitely	**Bare-Metal GPU Time-Slicing** Strict memory ceilings on RTX 5070 Ti

Promotion Lifecycle

Study Run → Candidate Model → Automated Conformal Gate → Staging Tag → Production Promotion

Audit Integrity

- **Cryptographic Binding:** Strict cryptographic binding connecting production model version to immutable DVC commit and signed Git release tag.
- **Safety Guarantee:** Prevents deployment of uncalibrated or drifting models.

Status: **[Implemented & Verified]**

Enterprise Level 4 Verification Matrix

- **Data Lineage:** 100% deterministic Parquet metadata with DVC SHA256 immutability.
- **Hermetic Toolchains:** Zero reliance on unpinned pip/conda environments; strict lockfiles via Pixi.
- **Mathematical Safety:** Automated test gates asserting conformal coverage $1 - \alpha$ on holdout splits.
- **Continuous Deployment & Observability:** Jenkins OCI image builds and k3s manifests established; streaming drift monitors and Label Studio triage scheduled as Phase 3 roadmap.

Status: **[Partially Implemented / In Progress]**

CHAPTER 10

Multimodal Knowledge Graph & Vision-RAG

Epistemic Ambiguity Resolution, Filtered Vector Retrieval & Clinical-Grade Diagnosis Cards

---

Status: [Target Architecture / Roadmap]

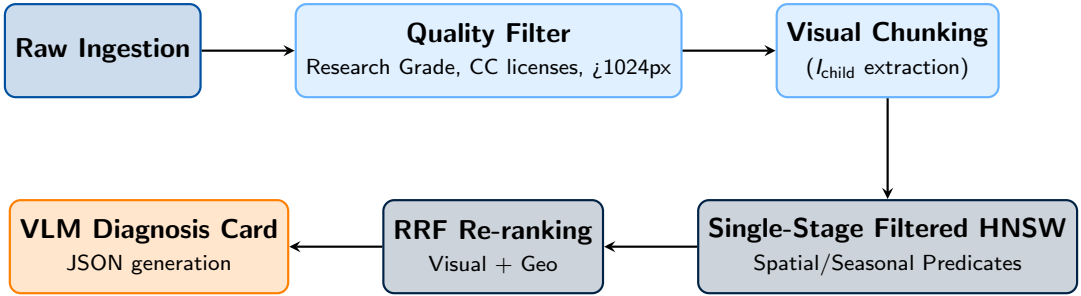
Conformal-Driven Lazy Invocation Policy

- **Throughput Preservation:** Querying dense graph and vector databases on routine inferences exhausts latency budgets.
- **Fast-Path Resolution:** Unambiguous singleton predictions ($|\hat{C}(X)| = 1$) resolve in ≤ 15 ms via INT8 ONNX models.
- **Lazy Activation Gate:** The MMKG pipeline triggers exclusively when the Split Conformal Gate detects epistemic uncertainty.
- **Epistemic Triggers:** Severe classification ambiguity ($|\hat{C}(X)| > 3$) between sibling taxa or complete OOD outliers ($|\hat{C}(X)| = 0$).

Status: [Target Architecture / Roadmap]

MMKG Pipeline Architecture

Ingestion & Retrieval Data Flow



Status: [Target Architecture / Roadmap]

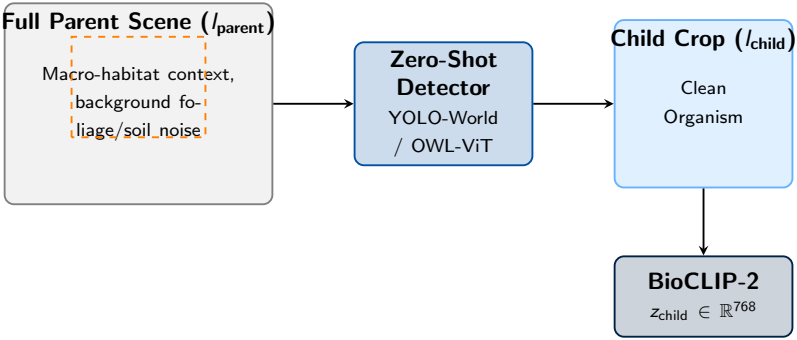
Background Clutter Isolation Strategy

- **The Foliage Noise Problem:** In uncropped citizen-science photos, specimens occupy $< 10\%$ of pixels; soil and sky corrupt embeddings.
- **Zero-Shot Organism Localization:** Upstream zero-shot detectors (YOLO-World / OWL-ViT) isolate primary organism bounding boxes.
- **Focused Child Embedding:** Extracts noise-free BioCLIP-2 embeddings ($z_{\text{child}} \in \mathbb{R}^{768}$) from the isolated specimen crop (I_{child}).
- **Context Preservation:** Uncropped parent scene context (I_{parent}) is retained for macro-habitat and environmental verification.

Status: [Target Architecture / Roadmap]

Visual Pre-Processing

Parent-Child Visual Chunking Diagram



Status: [Target Architecture / Roadmap]

Pruning Geographically Impossible Candidates

- **Convergent Evolution Trap:** Unrelated species on distant continents evolve near-identical morphologies (e.g., mimicry).
- **Post-Filtering Failure:** Top-k visual search post-filtering causes candidate starvation when top neighbors are from the wrong continent.
- **Native Single-Stage HNSW Walk:** Spatial predicates (Haversine geodesic $R_{\max} \leq 500$ km) evaluated directly inside C++ graph traversal.
- **Temporal Phenology Bounds:** Seasonal observation month filters ($\Delta_{\text{season}} = 1$) prune phenologically impossible taxa.

Status: [Target Architecture / Roadmap]

Synthesizing Visual Similarity & Ecological Priors

- **Rank Combination:** Combines dense visual cosine similarity ($w_{\text{vis}} = 0.65$) with GBIF historical occurrence density ($w_{\text{geo}} = 0.35$):

$$\text{RRF}(d) = \frac{w_{\text{vis}}}{k + r_{\text{vis}}(d)} + \frac{w_{\text{geo}}}{k + r_{\text{geo}}(d)}$$

- **Distribution Invariance:** Non-linear rank fusion prevents score scale mismatch between dissimilar probability spaces.
- **Robustness:** Constant $k = 60$ dampens top-rank outliers, tilting decisions toward true local ecology.

Status: [Target Architecture / Roadmap]

Grammar-Constrained VLM Reasoning for Triage

- **Taxonomist Context Synthesis:** When ambiguous observations hit human review, query crops and top-3 exemplar specimens are injected into a local VLM (Qwen2-VL-2B).
- **Grammar-Based Sampling:** Strict JSON grammar constraint prevents hallucination and guarantees valid schema adherence.
- **Differential Diagnosis Card:** Emits structured anatomical discriminators, likelihoods, and CAM activation overlays.
- **Label Studio Integration:** Direct rendering in review UI turns triage into rapid expert verification.

Status: [Target Architecture / Roadmap]

Handling Dynamic Biological Taxonomy

- **Dynamic Taxa Evolution:** Biological committees frequently revise classifications: merging species (lumps) or subdividing them (splits).
- **Decoupled Architecture:** Embeddings reside in HNSW vector storage, while taxonomic relationships and metadata reside in DuckDB.
- $\mathcal{O}(1)$ **Taxonomic Lumps:** Species mergers execute as instant relational table updates without touching model weights or embeddings.
- **Audited Taxonomic Splits:** Historical nodes are marked `requires_reannotation = true` and safely queued for expert re-labeling.

Status: [Target Architecture / Roadmap]

Conclusion

Core Architectural Principles & System Vision

The Pillars of Production Systems Engineering

TaxonVision is built on three uncompromising principles that define enterprise MLOps.

- **Reproducibility by Default:** Using Pixi environments, DVC data lineage, and declarative k3s manifests ensures every run is perfectly replicable.
- **Mathematical Safety under Uncertainty:** Split Conformal Prediction guarantees error bounds, refusing confident misclassification.
- **Hardware-Conscious Sustainability:** Bare-metal GPU time-slicing, INT8 ONNX quantization, and zero-allocation inference maximize throughput per watt.

Status: **[Implemented & Verified]**